In [1]:
```
# Project Assignment Part 6 follows after Project Assignment Part 5
```

In [2]:
```
# Project Assignment Part 2
```

In [3]:
```
# Revised Questions
# Research Question 1:
# Is there a statistically significant difference in average shipment volume (Boxes

# Research Question 2:
# Are there statistically significant differences in shipment volume (Boxes_Shipped
```

In [4]:
```
# Import and display data
import pandas as pd
df = pd.read_csv("Chocolate Sales (2).csv")

df.head()
df.info()
df.describe()
df.columns
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3282 entries, 0 to 3281
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Sales Person   3282 non-null   object
 1   Country        3282 non-null   object
 2   Product        3282 non-null   object
 3   Date           3282 non-null   object
 4   Amount         3282 non-null   object
 5   Boxes Shipped  3282 non-null   int64
dtypes: int64(1), object(5)
memory usage: 154.0+ KB
```

Out[4]:
```
Index(['Sales Person', 'Country', 'Product', 'Date', 'Amount',
       'Boxes Shipped'],
      dtype='object')
```

In [5]:
```
# Data Cleaning and Preparation
# Only keep 'Sales Person', 'Country', and 'Boxes Shipped', as they are related to
df = df[['Sales Person', 'Country', 'Boxes Shipped']].copy()

df
```

Out[5]:

	Sales Person	Country	Boxes Shipped
0	Jehu Rudeforth	UK	180
1	Van Tuxwell	India	94
2	Gigi Bohling	India	91
3	Jan Morforth	Australia	342
4	Jehu Rudeforth	UK	184
...	...	...	...
3277	Karlen McCaffrey	Australia	354
3278	Jehu Rudeforth	USA	121
3279	Ches Bonnell	Canada	238
3280	Dotty Strutley	India	397
3281	Karlen McCaffrey	India	355

3282 rows × 3 columns

In [6]:
```
# Handling Missing Data
df.isna().sum()
```

Out[6]:
```
Sales Person     0
Country          0
Boxes Shipped    0
dtype: int64
```

In [7]:
```
# There is no missing value in any of the columns, therefore, no value needs to be
```

In [8]:
```
# Data Transformations
```

In [9]:
```
# Re-label columns to ensure accurate process
df.columns = (
    df.columns
    .str.strip()
    .str.replace(" ", "_")
)

df.columns
```

Out[9]:
```
Index(['Sales_Person', 'Country', 'Boxes_Shipped'], dtype='object')
```

In [10]:
```
# Change column type
# Converting 'Sales_Person' and 'Country' to categorical makes the data structure a
# 'Boxes_Shipped' represents numerical shipment volume and must remain a numeric ty
df['Sales_Person'] = df['Sales_Person'].astype('category')
df['Country'] = df['Country'].astype('category')

df.dtypes
```

Out[10]:
```
Sales_Person     category
Country          category
Boxes_Shipped       int64
dtype: object
```

In [11]:
```
# Check for duplicated rows
any(df.duplicated())
```

Out[11]:
```
True
```

In [12]:
```
# Drop duplicated rows
df = df.drop_duplicates().copy()

df
```

Out[12]:

	Sales_Person	Country	Boxes_Shipped
0	Jehu Rudeforth	UK	180
1	Van Tuxwell	India	94
2	Gigi Bohling	India	91
3	Jan Morforth	Australia	342
4	Jehu Rudeforth	UK	184
...	...	...	...
3277	Karlen McCaffrey	Australia	354
3278	Jehu Rudeforth	USA	121
3279	Ches Bonnell	Canada	238
3280	Dotty Strutley	India	397
3281	Karlen McCaffrey	India	355

2920 rows × 3 columns

In [13]:
```
# Replace values
# No replacement needed for 'Sales_Person' or 'Boxes_Shipped'.
# Fix inconsistent country naming / typos if they exist. If they don't, nothing nee
df['Country'].cat.categories
```

Out[13]:
```
Index(['Australia', 'Canada', 'India', 'New Zealand', 'UK', 'USA'], dtype='objec
t')
```

In [14]:
```
# Detect and filter outliers
# IQR method
Q1 = df['Boxes_Shipped'].quantile(0.25)
Q3 = df['Boxes_Shipped'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
```

```
df['Outlier_Flag'] = (df['Boxes_Shipped'] < lower) | (df['Boxes_Shipped'] > upper)
df['Outlier_Flag'].value_counts()

# Unnecessary to remove outliers as the research question 2 is about shipment volum
```

Out[14]:
```
Outlier_Flag
False    2847
True       73
Name: count, dtype: int64
```

In [15]:
```
df.head()
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 2920 entries, 0 to 3281
Data columns (total 4 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Sales_Person   2920 non-null   category
 1   Country        2920 non-null   category
 2   Boxes_Shipped  2920 non-null   int64
 3   Outlier_Flag   2920 non-null   bool
dtypes: bool(1), category(2), int64(1)
memory usage: 55.1 KB
```

In [16]:
```
# Project Assignment Part 3
```

In [17]:
```
# Processing Strings
# Cleaned column names in part 2

# Clean text columns used for grouping in ANOVA
df['Country'] = df['Country'].astype(str).str.strip().str.title()
df['Sales_Person'] = df['Sales_Person'].astype(str).str.strip().str.title()

# Replace inconsistent labels if needed (edit mapping to match your dataset)
country_map = {
    "Usa": "USA",
    "U.S.": "USA",
    "United States": "USA",
    "Uk": "United Kingdom"
}
df['Country'] = df['Country'].replace(country_map)

# Quick check of unique labels (useful to confirm string cleaning worked)
df['Country'].unique(), df['Sales_Person'].unique()
```

Out[17]:
```
(array(['United Kingdom', 'India', 'Australia', 'New Zealand', 'USA',
        'Canada'], dtype=object),
 array(['Jehu Rudeforth', 'Van Tuxwell', 'Gigi Bohling', 'Jan Morforth',
        'Oby Sorrel', 'Gunar Cockshoot', 'Brien Boise',
        'Rafaelita Blaksland', 'Barr Faughny', 'Mallorie Waber',
        'Karlen Mccaffrey', "Marney O'Breen", 'Beverie Moffet',
        'Roddy Speechley', 'Curtice Advani', 'Husein Augar', 'Kaine Padly',
        'Dennison Crosswaite', "Wilone O'Kielt", 'Andria Kimpton',
        'Kelci Walkden', 'Camilla Castle', 'Madelene Upcott',
        'Dotty Strutley', 'Ches Bonnell'], dtype=object))
```

In [18]:
```
# Data Wrangling: Combining and Merging Datasets
# My project uses one dataset file, so there is no need to combine multiple files w

sp_summary = (
    df.groupby('Sales_Person', as_index=False)
      .agg(Total_Boxes=('Boxes_Shipped', 'sum'),
           Num_Shipments=('Boxes_Shipped', 'count'),
           Avg_Boxes=('Boxes_Shipped', 'mean'))
)

sp_summary.head()

# Merge the summary back into the main dataset
df = df.merge(sp_summary, on='Sales_Person', how='left')

df.head()
```

Out[18]:

	Sales_Person	Country	Boxes_Shipped	Outlier_Flag	Total_Boxes	Num_Shipments	Avg_
0	Jehu Rudeforth	United Kingdom	180	False	20944	119	176.0
1	Van Tuxwell	India	94	False	19118	129	148.2
2	Gigi Bohling	India	91	False	17663	123	143.6
3	Jan Morforth	Australia	342	False	22355	108	206.9
4	Jehu Rudeforth	United Kingdom	184	False	20944	119	176.0

In [19]:
```
# Merge the country summary
country_summary = (
    df.groupby('Country', as_index=False)
      .agg(Country_Total_Boxes=('Boxes_Shipped', 'sum'),
           Country_Num_Shipments=('Boxes_Shipped', 'count'),
           Country_Avg_Boxes=('Boxes_Shipped', 'mean'))
)

df = df.merge(country_summary, on='Country', how='left')

df.head()
```

Out[19]:

	Sales_Person	Country	Boxes_Shipped	Outlier_Flag	Total_Boxes	Num_Shipments	Avg_
0	Jehu Rudeforth	United Kingdom	180	False	20944	119	176.0
1	Van Tuxwell	India	94	False	19118	129	148.2
2	Gigi Bohling	India	91	False	17663	123	143.6
3	Jan Morforth	Australia	342	False	22355	108	206.9
4	Jehu Rudeforth	United Kingdom	184	False	20944	119	176.0

In [20]:

```
# Data Wrangling: Reshaping and Pivoting
# Average 'Boxes_Shipped' by 'Country' (table form)
country_pivot = df.pivot_table(
    index='Country',
    values='Boxes_Shipped',
    aggfunc='mean'
)

country_pivot.sort_values('Boxes_Shipped', ascending=False)
```

Out[20]:

Country	Boxes_Shipped
Canada	195.902386
United Kingdom	179.794979
Australia	172.119349
India	171.845070
New Zealand	166.801724
USA	164.160600

In [21]:

```
# Average 'Boxes_Shipped' by 'Sales_Person' (table form)
sales_pivot = df.pivot_table(
    index='Sales_Person',
    values='Boxes_Shipped',
    aggfunc='mean'
)

sales_pivot.sort_values('Boxes_Shipped', ascending=False).head(10)
```

Out[21]:

Sales_Person	Boxes_Shipped
Karlen Mccaffrey	220.742188
Jan Morforth	206.990741
Dennison Crosswaite	201.126984
Dotty Strutley	200.680000
Marney O'Breen	197.181034
Beverie Moffet	193.433824
Oby Sorrel	190.244275
Camilla Castle	183.863636
Andria Kimpton	180.084906
Madelene Upcott	177.415254

In [22]:

```
# Make a two-level index, then unstack to a table
frame = df[['Country', 'Sales_Person', 'Boxes_Shipped']].groupby(['Country', 'Sales
frame2 = frame.unstack(level='Sales_Person')

frame2.head()
```

Out[22]:

Sales_Person	Andria Kimpton	Barr Faughny	Beverie Moffet	Brien Boise	Camilla Castle	Ches Bonnell	Curtice Advani	Dennison Crosswaite	St
Country									
Australia	3973	1842	6563	4329	3270	2770	3090	4552	
Canada	5012	3839	3766	4982	1277	3950	2021	3676	
India	2610	2789	4817	4571	3393	3922	4539	2726	
New Zealand	2944	4792	3414	4024	2577	3962	4514	4250	
USA	1455	2594	4436	3432	3770	2373	3162	5821	

5 rows × 25 columns

In [23]:

```
# Convert the table back into long format (like lecture melt)
frame2_reset = frame2.reset_index()
frame2_reset.columns

# Flatten multi-level column names
frame2_reset.columns = [
```

```
    col[0] if col[1] == '' else col[1]
    for col in frame2_reset.columns
]

frame2_reset.head()

melted = pd.melt(
    frame2_reset,
    id_vars='Country',
    var_name='Sales_Person',
    value_name='Boxes_Shipped'
)

melted
```

Out[23]:

	Country	Sales_Person	Boxes_Shipped
0	Australia	Andria Kimpton	3973
1	Canada	Andria Kimpton	5012
2	India	Andria Kimpton	2610
3	New Zealand	Andria Kimpton	2944
4	USA	Andria Kimpton	1455
...	...	...	...
145	Canada	Wilone O'Kielt	1053
146	India	Wilone O'Kielt	2618
147	New Zealand	Wilone O'Kielt	3623
148	USA	Wilone O'Kielt	1224
149	United Kingdom	Wilone O'Kielt	469

150 rows × 3 columns

In [24]:

```
# Project Assignment Part 4
# Plotting and Visualization
```

In [25]:

```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

In [26]:

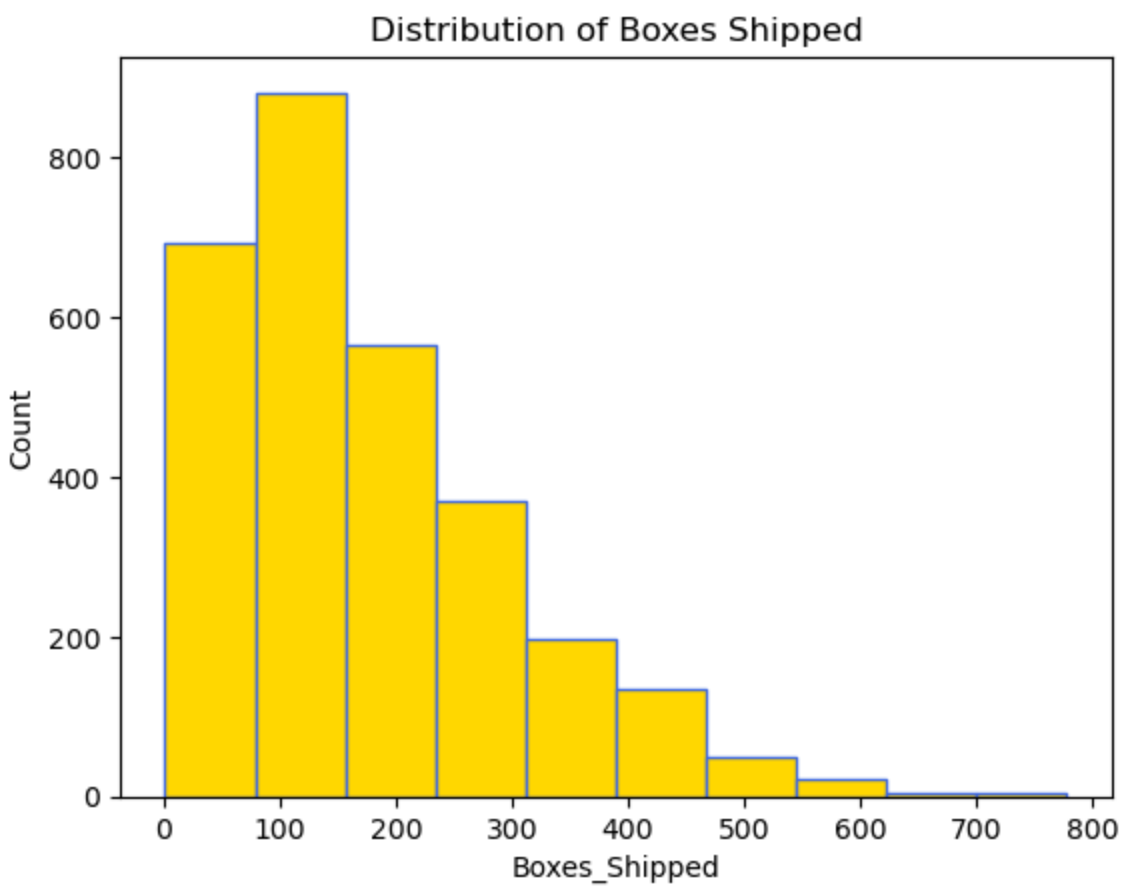
```
df[['Boxes_Shipped']].describe()
```

Out[26]:

	Boxes_Shipped
count	2920.000000
mean	174.966096
std	124.402767
min	1.000000
25%	82.000000
50%	145.000000
75%	243.000000
max	778.000000

In [27]:

```
# Histogram of Boxes_Shipped (explore distribution + outliers)
plt.hist(df['Boxes_Shipped'], color='gold', edgecolor='royalblue')
plt.title('Distribution of Boxes Shipped')
plt.xlabel('Boxes_Shipped')
plt.ylabel('Count')
plt.show()
```

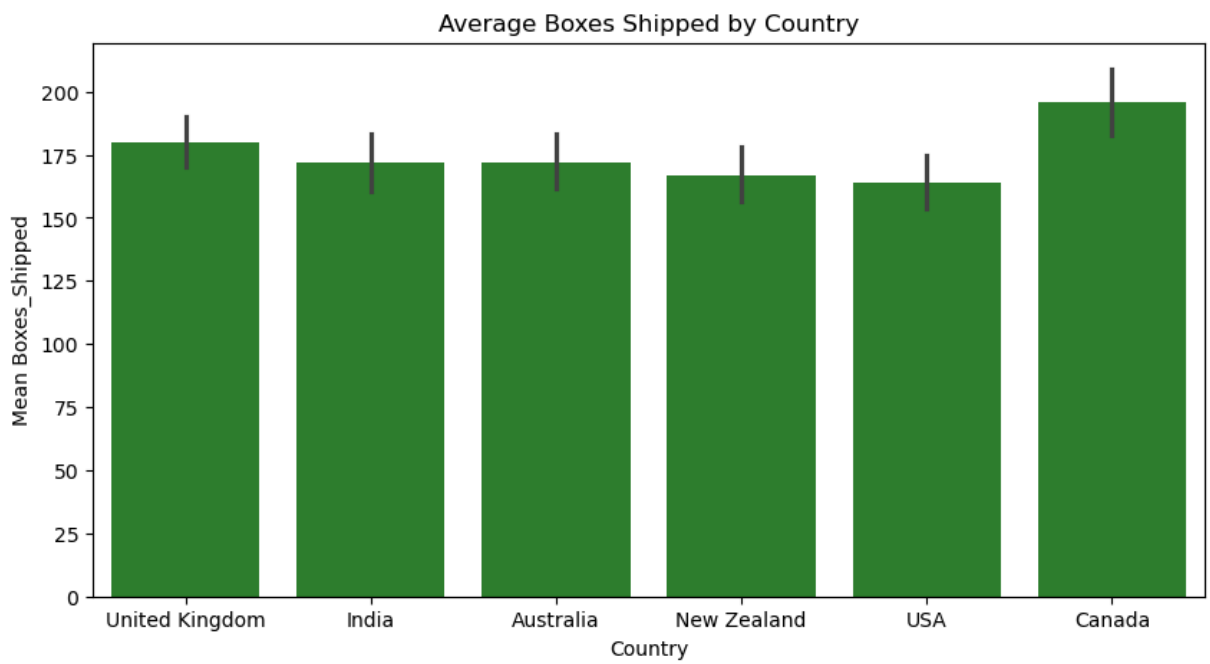


In [28]:

```
# Boxplot: Boxes_Shipped by Country (explore differences for RQ2)
plt.figure(figsize=(10, 5))
sns.boxplot(data=df, x='Country', y='Boxes_Shipped', color='lightpink')
plt.title('Boxes Shipped by Country')
```

```
plt.xlabel('Country')
plt.ylabel('Boxes_Shipped')
plt.show()
```

In [29]:
```
# Barplot: Mean Boxes_Shipped by Country (explore average differences for RQ2)
plt.figure(figsize=(10, 5))
sns.barplot(data=df, x='Country', y='Boxes_Shipped', estimator=np.mean, color='fore
plt.title('Average Boxes Shipped by Country')
plt.xlabel('Country')
plt.ylabel('Mean Boxes_Shipped')
plt.show()
```

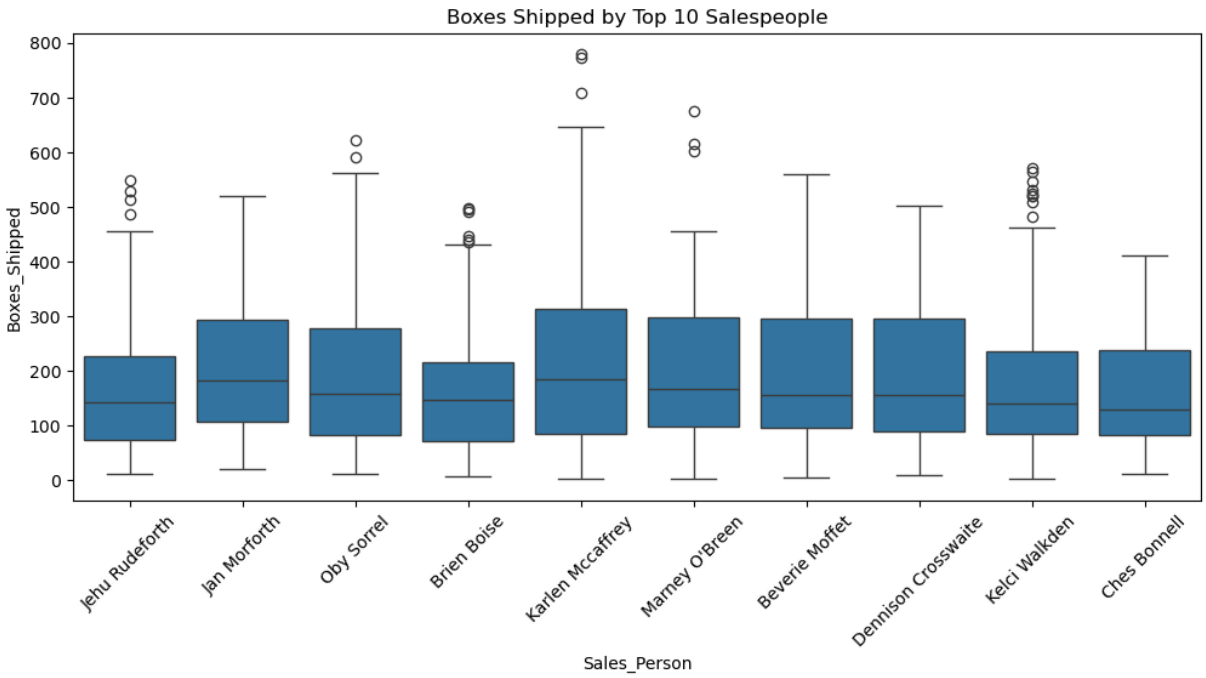


In [30]:
```
# Top 10 salespeople (exploratory setup for RQ1) + boxplot/barplot
# Identify top 10 by total shipped
top10 = (
```

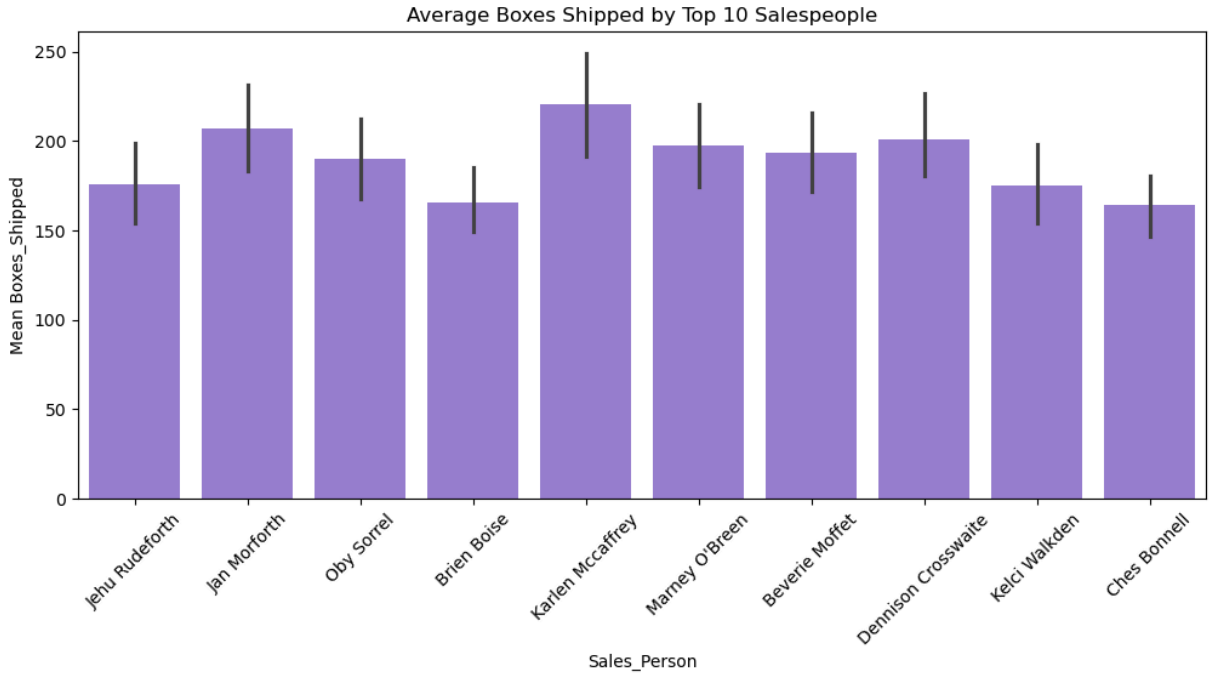
```
    df.groupby('Sales_Person')['Boxes_Shipped']
      .sum()
      .sort_values(ascending=False)
      .head(10)
      .index
)

df_top10 = df[df['Sales_Person'].isin(top10)].copy()
df_top10['Sales_Person'].nunique()
```

Out[30]: 10

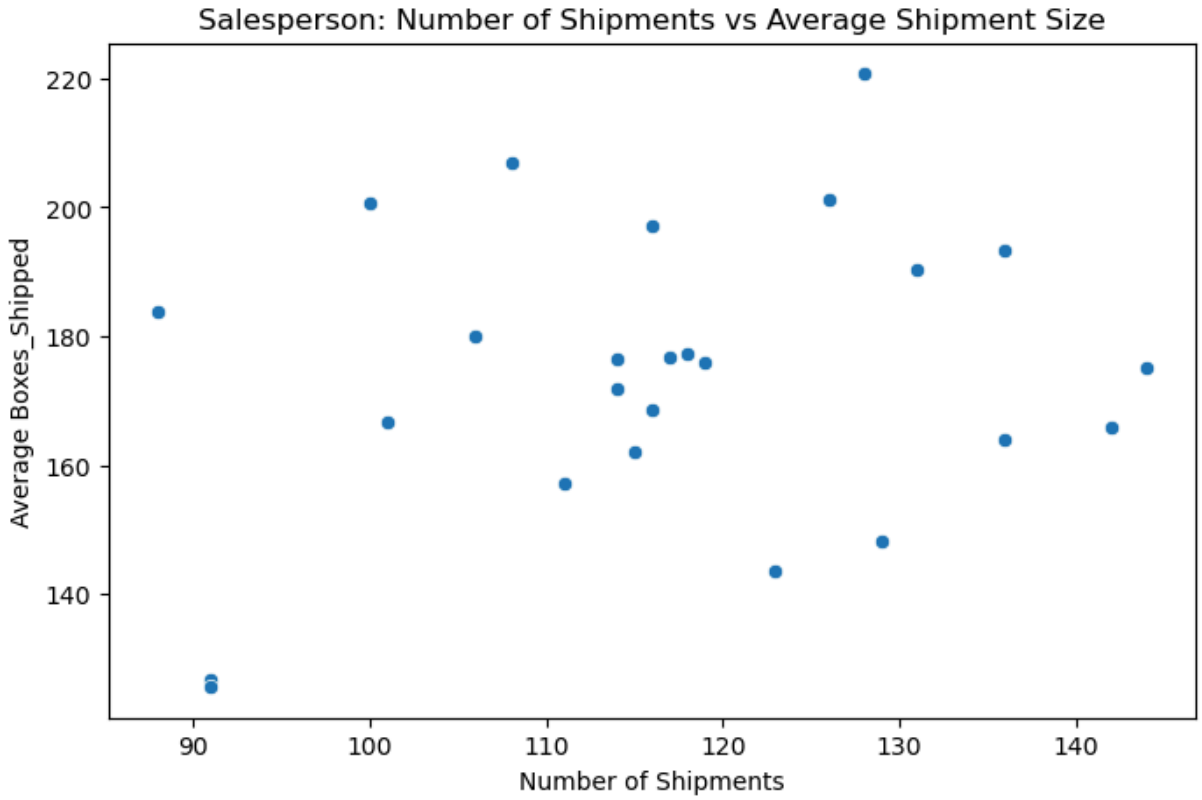
In [31]:
```
# Boxplot: Boxes_Shipped by Top 10 Salespeople
plt.figure(figsize=(12, 5))
sns.boxplot(data=df_top10, x='Sales_Person', y='Boxes_Shipped')
plt.title('Boxes Shipped by Top 10 Salespeople')
plt.xlabel('Sales_Person')
plt.ylabel('Boxes_Shipped')
plt.xticks(rotation=45)
plt.show()
```



In [32]:
```
# Barplot: Mean Boxes_Shipped by Top 10 Salespeople
plt.figure(figsize=(12, 5))
sns.barplot(data=df_top10, x='Sales_Person', y='Boxes_Shipped', estimator=np.mean,
plt.title('Average Boxes Shipped by Top 10 Salespeople')
plt.xlabel('Sales_Person')
plt.ylabel('Mean Boxes_Shipped')
plt.xticks(rotation=45)
plt.show()
```



In [33]:
```
# Scatterplot: Total shipments vs average shipment size by salesperson
# This is exploratory and helps interpret “high performer” patterns beyond just one
sp_summary = (
    df.groupby('Sales_Person', as_index=False)
      .agg(Total_Boxes=('Boxes_Shipped', 'sum'),
           Avg_Boxes=('Boxes_Shipped', 'mean'),
           Num_Shipments=('Boxes_Shipped', 'count'))
)

plt.figure(figsize=(8, 5))
sns.scatterplot(data=sp_summary, x='Num_Shipments', y='Avg_Boxes')
plt.title('Salesperson: Number of Shipments vs Average Shipment Size')
plt.xlabel('Number of Shipments')
plt.ylabel('Average Boxes_Shipped')
plt.show()
```



In [34]:
```
# Project Assignment 5
```

In [35]:
```
# 1. Data Aggregation and Group Operations
```

In [36]:
```
# Group Boxes_Shipped by Country
grouped_country = df["Boxes_Shipped"].groupby(df["Country"])
grouped_country
```

Out[36]:
```
<pandas.core.groupby.generic.SeriesGroupBy object at 0x000002920969ABD0>
```

In [37]:
```
# Mean shipment volume for each country
grouped_country.mean()
```

Out[37]:
```
Country
Australia         172.119349
Canada            195.902386
India             171.845070
New Zealand       166.801724
USA               164.160600
United Kingdom    179.794979
Name: Boxes_Shipped, dtype: float64
```

In [38]:
```
# Group the full dataframe by Country
all_grouped_country = df.groupby("Country")
all_grouped_country.mean(numeric_only=True)
```

Out[38]:

Country	Boxes_Shipped	Outlier_Flag	Total_Boxes	Num_Shipments	Avg_Boxes	Country
Australia	172.119349	0.018083	20444.600362	117.195298	173.693480	
Canada	195.902386	0.056399	21149.731020	119.737527	176.112644	
India	171.845070	0.034205	21158.496982	119.818913	176.185553	
New Zealand	166.801724	0.019397	20664.816810	118.187500	174.075562	
USA	164.160600	0.008565	20769.693790	118.269807	174.976813	
United Kingdom	179.794979	0.014644	20926.838912	119.424686	174.918671	

In [39]:
```
# Count the number of rows in each country group
all_grouped_country.size()
```

Out[39]:
```
Country
Australia         553
Canada            461
India             497
New Zealand       464
USA               467
United Kingdom    478
dtype: int64
```

In [40]:
```
# 2.1 Grouping Data
```

In [41]:
```
means = df["Boxes_Shipped"].groupby([df["Country"], df["Sales_Person"]]).mean()
means
```

Out[41]:
```
Country         Sales_Person
Australia       Andria Kimpton         198.650000
                Barr Faughny           122.800000
                Beverie Moffet         312.523810
                Brien Boise            144.300000
                Camilla Castle         181.666667
                                          ...
United Kingdom  Oby Sorrel             214.827586
                Rafaelita Blaksland     97.384615
                Roddy Speechley        264.272727
                Van Tuxwell            126.521739
                Wilone O'Kielt          58.625000
Name: Boxes_Shipped, Length: 150, dtype: float64
```

In [42]:
```
means.unstack()
```

Out[42]:

Sales_Person Country	Andria Kimpton	Barr Faughny	Beverie Moffet	Brien Boise	Camilla Castle	Ches Bonnell	C A
Australia	198.650000	122.800000	312.523810	144.300000	181.666667	131.904762	134.3
Canada	200.480000	182.809524	163.739130	237.238095	159.625000	232.352941	106.3
India	153.529412	199.214286	172.035714	163.250000	226.200000	135.241379	189.1
New Zealand	173.176471	159.733333	148.434783	182.909091	135.631579	152.384615	250.7
USA	132.272727	117.909091	233.473684	163.428571	198.421053	139.588235	225.8
United Kingdom	193.437500	213.000000	150.500000	110.700000	210.333333	204.269231	141.5

6 rows × 25 columns

In [43]:

```
# Mean and standard deviation of shipment volume by Country
grouped_country.agg(["mean", "std"])
```

Out[43]:

Country	mean	std
Australia	172.119349	129.603297
Canada	195.902386	142.983372
India	171.845070	125.127508
New Zealand	166.801724	120.087853
USA	164.160600	111.428656
United Kingdom	179.794979	111.793891

In [44]:

```
# Custom labels like lecture
ftuples = [("Average", "mean"), ("Standard Deviation", "std")]
grouped_country.agg(ftuples)
```

Out[44]:

Country	Average	Standard Deviation
Australia	172.119349	129.603297
Canada	195.902386	142.983372
India	171.845070	125.127508
New Zealand	166.801724	120.087853
USA	164.160600	111.428656
United Kingdom	179.794979	111.793891

In [45]:

```
# 2.2 Grouping Salespeople for Research Question 1
# Find total boxes shipped by salesperson
sales_totals = df.groupby("Sales_Person")["Boxes_Shipped"].sum().sort_values(ascend
sales_totals
```

Out[45]:

```
Sales_Person
Karlen Mccaffrey       28255
Beverie Moffet         26307
Dennison Crosswaite    25342
Kelci Walkden          25205
Oby Sorrel             24922
Brien Boise            23552
Marney O'Breen         22873
Jan Morforth           22355
Ches Bonnell           22288
Jehu Rudeforth         20944
Madelene Upcott        20935
Kaine Padly            20674
Roddy Speechley        20110
Dotty Strutley         20068
Curtice Advani         19591
Gunar Cockshoot        19567
Van Tuxwell            19118
Andria Kimpton         19089
Barr Faughny           18625
Gigi Bohling           17663
Mallorie Waber         17431
Husein Augar           16846
Camilla Castle         16180
Rafaelita Blaksland    11535
Wilone O'Kielt         11426
Name: Boxes_Shipped, dtype: int64
```

In [46]:

```
# Select top 10 salespeople
top10_sales = sales_totals.head(10).index
top10_sales
```

Out[46]:
```
Index(['Karlen Mccaffrey', 'Beverie Moffet', 'Dennison Crosswaite',
       'Kelci Walkden', 'Oby Sorrel', 'Brien Boise', 'Marney O'Breen',
       'Jan Morforth', 'Ches Bonnell', 'Jehu Rudeforth'],
      dtype='object', name='Sales_Person')
```

In [47]:
```
# Filter the dataframe to keep only top 10 salespeople
df_top10 = df[df["Sales_Person"].isin(top10_sales)].copy()
df_top10.head()
```

Out[47]:

	Sales_Person	Country	Boxes_Shipped	Outlier_Flag	Total_Boxes	Num_Shipments	Avg_
0	Jehu Rudeforth	United Kingdom	180	False	20944	119	176.0
3	Jan Morforth	Australia	342	False	22355	108	206.9
4	Jehu Rudeforth	United Kingdom	184	False	20944	119	176.0
6	Oby Sorrel	United Kingdom	176	False	24922	131	190.2
8	Jehu Rudeforth	New Zealand	59	False	20944	119	176.0

In [48]:
```
# Group shipment volume by top 10 salespeople
grouped_sales = df_top10["Boxes_Shipped"].groupby(df_top10["Sales_Person"])
grouped_sales.mean()
```

Out[48]:
```
Sales_Person
Beverie Moffet         193.433824
Brien Boise            165.859155
Ches Bonnell           163.882353
Dennison Crosswaite    201.126984
Jan Morforth           206.990741
Jehu Rudeforth         176.000000
Karlen Mccaffrey       220.742188
Kelci Walkden          175.034722
Marney O'Breen         197.181034
Oby Sorrel             190.244275
Name: Boxes_Shipped, dtype: float64
```

In [49]:
```
# Summary statistics for top 10 salespeople
grouped_sales.agg(["mean", "std", "count"])
```

Out[49]:

Sales_Person	mean	std	count
Beverie Moffet	193.433824	132.920567	136
Brien Boise	165.859155	114.663714	142
Ches Bonnell	163.882353	104.711601	136
Dennison Crosswaite	201.126984	135.671838	126
Jan Morforth	206.990741	128.020186	108
Jehu Rudeforth	176.000000	127.133521	119
Karlen Mccaffrey	220.742188	162.318257	128
Kelci Walkden	175.034722	130.364117	144
Marney O'Breen	197.181034	135.318664	116
Oby Sorrel	190.244275	131.210230	131

In [50]:

```
# 2.3 Apply Function to Grouped Data
# Define a function that returns the top n shipments by Boxes_Shipped
def top(df, n=5, column="Boxes_Shipped"):
    return df.sort_values(column, ascending=False)[:n]

top(df, n=5)
```

Out[50]:

	Sales_Person	Country	Boxes_Shipped	Outlier_Flag	Total_Boxes	Num_Shipments	A
2625	Karlen Mccaffrey	Australia	778	True	28255	128	2
1721	Karlen Mccaffrey	Australia	772	True	28255	128	2
1986	Van Tuxwell	Canada	746	True	19118	129	1
1016	Van Tuxwell	Canada	709	True	19118	129	1
725	Karlen Mccaffrey	Australia	708	True	28255	128	2

In [51]:

```
# Top 3 shipments for each country
df.groupby("Country").apply(top, n=3, column="Boxes_Shipped")
```

```
C:\Users\Colleen\AppData\Local\Temp\ipykernel_24912\1853698895.py:2: FutureWarning:
DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecate
d, and in a future version of pandas the grouping columns will be excluded from the
operation. Either pass `include_groups=False` to exclude the groupings or explicitly
select the grouping columns after groupby to silence this warning.
  df.groupby("Country").apply(top, n=3, column="Boxes_Shipped")
```

Out[51]:

Country		Sales_Person	Country	Boxes_Shipped	Outlier_Flag	Total_Boxes	Num_Sh
Australia	**2625**	Karlen Mccaffrey	Australia	778	True	28255	
	1721	Karlen Mccaffrey	Australia	772	True	28255	
	725	Karlen Mccaffrey	Australia	708	True	28255	
Canada	**1986**	Van Tuxwell	Canada	746	True	19118	
	1016	Van Tuxwell	Canada	709	True	19118	
	2866	Van Tuxwell	Canada	679	True	19118	
India	**2075**	Karlen Mccaffrey	India	605	True	28255	
	1117	Karlen Mccaffrey	India	583	True	28255	
	39	Karlen Mccaffrey	India	581	True	28255	
New Zealand	**2743**	Beverie Moffet	New Zealand	560	True	26307	
	2735	Kaine Padly	New Zealand	548	True	20674	
	855	Kaine Padly	New Zealand	518	True	20674	
USA	**2595**	Dotty Strutley	USA	538	True	20068	
	1687	Dotty Strutley	USA	536	True	20068	
	687	Dotty Strutley	USA	508	True	20068	
United Kingdom	**1379**	Kaine Padly	United Kingdom	599	True	20674	
	2305	Kaine Padly	United Kingdom	594	True	20674	
	337	Kaine Padly	United Kingdom	554	True	20674	

In [52]:

```
# Top 2 shipments for each top-10 salesperson
df_top10.groupby("Sales_Person").apply(top, n=2, column="Boxes_Shipped")
```

```
C:\Users\Colleen\AppData\Local\Temp\ipykernel_24912\776346292.py:2: FutureWarning: D
ataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated,
and in a future version of pandas the grouping columns will be excluded from the ope
ration. Either pass `include_groups=False` to exclude the groupings or explicitly se
lect the grouping columns after groupby to silence this warning.
  df_top10.groupby("Sales_Person").apply(top, n=2, column="Boxes_Shipped")
```

Out[52]:

Sales_Person		Sales_Person	Country	Boxes_Shipped	Outlier_Flag	Total_Boxes	Num
Beverie Moffet	**2743**	Beverie Moffet	New Zealand	560	True	26307	
	865	Beverie Moffet	New Zealand	518	True	26307	
Brien Boise	**1800**	Brien Boise	Canada	497	True	23552	
	814	Brien Boise	Canada	495	True	23552	
Ches Bonnell	**1374**	Ches Bonnell	United Kingdom	411	False	22288	
	2301	Ches Bonnell	United Kingdom	406	False	22288	
Dennison Crosswaite	**1997**	Dennison Crosswaite	USA	501	True	25342	
	1590	Dennison Crosswaite	Canada	488	True	25342	
Jan Morforth	**109**	Jan Morforth	India	520	True	22355	
	1835	Jan Morforth	United Kingdom	507	True	22355	
Jehu Rudeforth	**1670**	Jehu Rudeforth	Canada	548	True	20944	
	2577	Jehu Rudeforth	Canada	529	True	20944	
Karlen Mccaffrey	**2625**	Karlen Mccaffrey	Australia	778	True	28255	
	1721	Karlen Mccaffrey	Australia	772	True	28255	
Kelci Walkden	**1185**	Kelci Walkden	Australia	570	True	25205	
	2171	Kelci Walkden	India	563	True	25205	
Marney O'Breen	**2725**	Marney O'Breen	Canada	674	True	22873	
	843	Marney O'Breen	Canada	614	True	22873	
Oby Sorrel	**2900**	Oby Sorrel	Canada	621	True	24922	
	1058	Oby Sorrel	Canada	591	True	24922	

In [53]:
```
# 3. Pivot Tables and Cross-Tabulation
```

In [54]:
```
# Pivot table: average shipment volume by Country
df.pivot_table(index="Country", values="Boxes_Shipped")
```

Out[54]:

Country	Boxes_Shipped
Australia	172.119349
Canada	195.902386
India	171.845070
New Zealand	166.801724
USA	164.160600
United Kingdom	179.794979

In [55]:
```
# Pivot table: mean and standard deviation by Country
df.pivot_table(index="Country",
               values="Boxes_Shipped",
               aggfunc=["mean", "std"])
```

Out[55]:

	mean	std
	Boxes_Shipped	Boxes_Shipped
Country		
Australia	172.119349	129.603297
Canada	195.902386	142.983372
India	171.845070	125.127508
New Zealand	166.801724	120.087853
USA	164.160600	111.428656
United Kingdom	179.794979	111.793891

In [56]:
```
# Pivot table: average shipment volume by Country and Sales_Person
df.pivot_table(index=["Country", "Sales_Person"],
               values="Boxes_Shipped")
```

Out[56]:

Country	Sales_Person	Boxes_Shipped
Australia	Andria Kimpton	198.650000
	Barr Faughny	122.800000
	Beverie Moffet	312.523810
	Brien Boise	144.300000
	Camilla Castle	181.666667
...	...	...
United Kingdom	Oby Sorrel	214.827586
	Rafaelita Blaksland	97.384615
	Roddy Speechley	264.272727
	Van Tuxwell	126.521739
	Wilone O'Kielt	58.625000

150 rows × 1 columns

In [57]:

```
# Pivot table for top 10 salespeople only
df_top10.pivot_table(index="Sales_Person",
                     values="Boxes_Shipped",
                     aggfunc=["mean", "std", "count"])
```

Out[57]:

	mean	std	count
	Boxes_Shipped	Boxes_Shipped	Boxes_Shipped
Sales_Person			
Beverie Moffet	193.433824	132.920567	136
Brien Boise	165.859155	114.663714	142
Ches Bonnell	163.882353	104.711601	136
Dennison Crosswaite	201.126984	135.671838	126
Jan Morforth	206.990741	128.020186	108
Jehu Rudeforth	176.000000	127.133521	119
Karlen Mccaffrey	220.742188	162.318257	128
Kelci Walkden	175.034722	130.364117	144
Marney O'Breen	197.181034	135.318664	116
Oby Sorrel	190.244275	131.210230	131

In [58]:
```
# Cross-Tabulation: frequency of shipments by Country
pd.crosstab(df["Country"], columns="Count")
```

Out[58]:

col_0	Count
Country	
Australia	553
Canada	461
India	497
New Zealand	464
USA	467
United Kingdom	478

In [59]:
```
# Cross-Tabulation: frequency of shipments by Sales Person
pd.crosstab(df["Sales_Person"], columns="Count")
```

Out[59]:

col_0	Count
Sales_Person	
Andria Kimpton	106
Barr Faughny	115
Beverie Moffet	136
Brien Boise	142
Camilla Castle	88
Ches Bonnell	136
Curtice Advani	114
Dennison Crosswaite	126
Dotty Strutley	100
Gigi Bohling	123
Gunar Cockshoot	116
Husein Augar	101
Jan Morforth	108
Jehu Rudeforth	119
Kaine Padly	117
Karlen Mccaffrey	128
Kelci Walkden	144
Madelene Upcott	118
Mallorie Waber	111
Marney O'Breen	116
Oby Sorrel	131
Rafaelita Blaksland	91
Roddy Speechley	114
Van Tuxwell	129
Wilone O'Kielt	91

In [60]:

```
# Cross-Tabulation: country by Sales Person
pd.crosstab(df["Country"], df["Sales_Person"])
```

Out[60]:

Sales_Person Country	Andria Kimpton	Barr Faughny	Beverie Moffet	Brien Boise	Camilla Castle	Ches Bonnell	Curtice Advani	Dennison Crosswaite	St
Australia	20	15	21	30	18	21	23	22	
Canada	25	21	23	21	8	17	19	19	
India	17	14	28	28	15	29	24	20	
New Zealand	17	30	23	22	19	26	18	17	
USA	11	22	19	21	19	17	14	30	
United Kingdom	16	13	22	20	9	26	16	18	

6 rows × 25 columns

In [61]:

```
# Cross-Tabulation: Totals
pd.crosstab(df["Country"], df["Sales_Person"], margins=True)
```

Out[61]:

Sales_Person Country	Andria Kimpton	Barr Faughny	Beverie Moffet	Brien Boise	Camilla Castle	Ches Bonnell	Curtice Advani	Dennison Crosswaite	St
Australia	20	15	21	30	18	21	23	22	
Canada	25	21	23	21	8	17	19	19	
India	17	14	28	28	15	29	24	20	
New Zealand	17	30	23	22	19	26	18	17	
USA	11	22	19	21	19	17	14	30	
United Kingdom	16	13	22	20	9	26	16	18	
All	106	115	136	142	88	136	114	126	

7 rows × 26 columns

In [62]:

```
df.head()
df_top10.head()
```

Out[62]:

	Sales_Person	Country	Boxes_Shipped	Outlier_Flag	Total_Boxes	Num_Shipments	Avg_
0	Jehu Rudeforth	United Kingdom	180	False	20944	119	176.0
3	Jan Morforth	Australia	342	False	22355	108	206.9
4	Jehu Rudeforth	United Kingdom	184	False	20944	119	176.0
6	Oby Sorrel	United Kingdom	176	False	24922	131	190.2
8	Jehu Rudeforth	New Zealand	59	False	20944	119	176.0

In [63]:
```
# Project Assignment Part 6
```

In [64]:
```
# Descriptive Statistics
df['Boxes_Shipped'].describe()
```

Out[64]:
```
count    2920.000000
mean      174.966096
std       124.402767
min         1.000000
25%        82.000000
50%       145.000000
75%       243.000000
max       778.000000
Name: Boxes_Shipped, dtype: float64
```

In [65]:
```
# Average shipment volume by country
df.groupby("Country")["Boxes_Shipped"].mean()
```

Out[65]:
```
Country
Australia         172.119349
Canada            195.902386
India             171.845070
New Zealand       166.801724
USA               164.160600
United Kingdom    179.794979
Name: Boxes_Shipped, dtype: float64
```

In [66]:
```
# Average shipment volume by salesperson
df.groupby("Sales_Person")["Boxes_Shipped"].mean()
```

Out[66]:
```
Sales_Person
Andria Kimpton         180.084906
Barr Faughny           161.956522
Beverie Moffet         193.433824
Brien Boise            165.859155
Camilla Castle         183.863636
Ches Bonnell           163.882353
Curtice Advani         171.850877
Dennison Crosswaite    201.126984
Dotty Strutley         200.680000
Gigi Bohling           143.601626
Gunar Cockshoot        168.681034
Husein Augar           166.792079
Jan Morforth           206.990741
Jehu Rudeforth         176.000000
Kaine Padly            176.700855
Karlen Mccaffrey       220.742188
Kelci Walkden          175.034722
Madelene Upcott        177.415254
Mallorie Waber         157.036036
Marney O'Breen         197.181034
Oby Sorrel             190.244275
Rafaelita Blaksland    126.758242
Roddy Speechley        176.403509
Van Tuxwell            148.201550
Wilone O'Kielt         125.560440
Name: Boxes_Shipped, dtype: float64
```

In [67]:
```
# ANOVA – Shipment Volume by Country - Research Question 2
from scipy import stats
groups_country = []

for name, group in df.groupby("Country"):
    groups_country.append(group["Boxes_Shipped"])

anova_country = stats.f_oneway(*groups_country)

anova_country
```

Out[67]:
```
F_onewayResult(statistic=np.float64(4.0007149993275375), pvalue=np.float64(0.00127
9513189023862))
```

In [68]:
```
# A one-way ANOVA test was conducted to examine whether the average shipment volume
```

In [69]:
```
# Identify Top 10 Salespeople
top10_sales = (
    df.groupby("Sales_Person")["Boxes_Shipped"]
      .sum()
      .sort_values(ascending=False)
      .head(10)
      .index
)

df_top10 = df[df["Sales_Person"].isin(top10_sales)].copy()
```

```
top10_sales
```

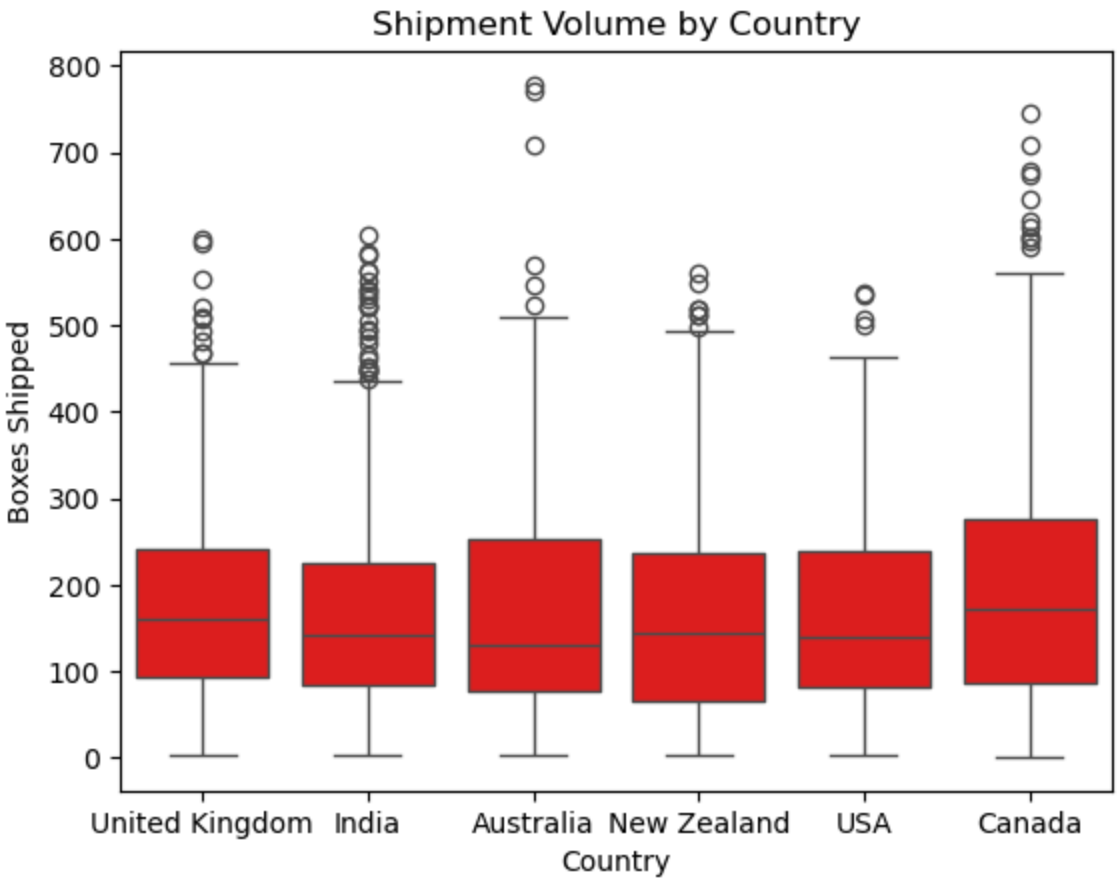
Out[69]:
```
Index(['Karlen Mccaffrey', 'Beverie Moffet', 'Dennison Crosswaite',
       'Kelci Walkden', 'Oby Sorrel', 'Brien Boise', 'Marney O'Breen',
       'Jan Morforth', 'Ches Bonnell', 'Jehu Rudeforth'],
      dtype='object', name='Sales_Person')
```

In [70]:
```
# ANOVA – Shipment Volume by Top 10 Salespeople - Research Question 1
groups_sales = []

for name, group in df_top10.groupby("Sales_Person"):
    groups_sales.append(group["Boxes_Shipped"])

anova_sales = stats.f_oneway(*groups_sales)

anova_sales
```

Out[70]:
```
F_onewayResult(statistic=np.float64(2.616491837324008), pvalue=np.float64(0.005362
4378441752794))
```

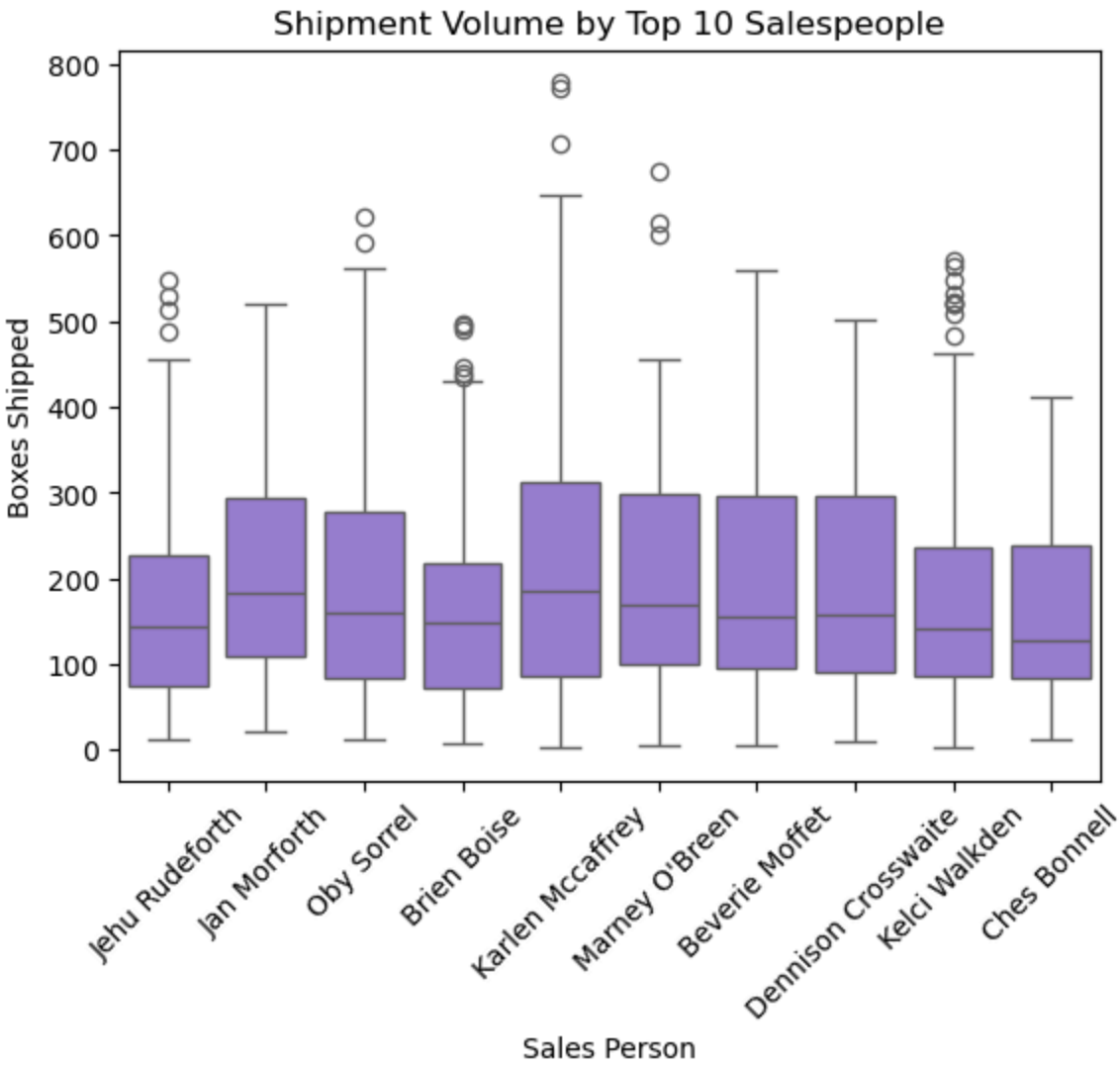
In [71]:
```
# A one-way ANOVA test was conducted to examine whether the average shipment volume
```

In [72]:
```
# Visualization of ANOVA Groups

# Shipment volume by country
sns.boxplot(data=df, x="Country", y="Boxes_Shipped", color="red")

plt.title("Shipment Volume by Country")
plt.xlabel("Country")
plt.ylabel("Boxes Shipped")

plt.show()
```



In [73]:
```
# The boxplot illustrates the distribution of shipment volume (Boxes_Shipped) acros
```

In [74]:
```
# Shipment volume by top 10 salespeople
sns.boxplot(data=df_top10, x="Sales_Person", y="Boxes_Shipped", color="mediumpurple

plt.xticks(rotation=45)

plt.title("Shipment Volume by Top 10 Salespeople")
plt.xlabel("Sales Person")
plt.ylabel("Boxes Shipped")

plt.show()
```



In [75]:
```
# The boxplot shows the distribution of shipment volume (Boxes_Shipped) among the t
```

In [76]:
```
# Post-Hoc Test (Tukey HSD)
from statsmodels.stats.multicomp import pairwise_tukeyhsd
```

In [77]:
```
# Tukey test for country
tukey_country = pairwise_tukeyhsd(
    endog=df["Boxes_Shipped"],
    groups=df["Country"],
    alpha=0.05
)

print(tukey_country)
```

```
      Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================================
   group1       group2     meandiff p-adj   lower    upper  reject
---------------------------------------------------------------------
  Australia         Canada   23.783 0.0289   1.4672 46.0988   True
  Australia          India  -0.2743    1.0 -22.1449 21.5963  False
  Australia    New Zealand  -5.3176 0.9841  -27.594 16.9588  False
  Australia            USA  -7.9587 0.9112 -30.1962 14.2787  False
  Australia United Kingdom   7.6756 0.9211 -14.4227 29.7739  False
     Canada          India -24.0573 0.0327 -46.9376 -1.1771   True
     Canada    New Zealand -29.1007 0.0049 -52.3691 -5.8322   True
     Canada            USA -31.7418 0.0014  -54.973 -8.5106   True
     Canada United Kingdom -16.1074 0.3491 -39.2054  6.9906  False
      India    New Zealand  -5.0433 0.9888 -27.8852 17.7985  False
      India            USA  -7.6845 0.9301 -30.4883 15.1194  False
      India United Kingdom   7.9499  0.918 -14.7183 30.6181  False
New Zealand            USA  -2.6411 0.9995 -25.8345 20.5522  False
New Zealand United Kingdom  12.9933 0.5941 -10.0667 36.0532  False
        USA United Kingdom  15.6344 0.3799   -7.388 38.6567  False
---------------------------------------------------------------------
```

In [78]:

```
# A Tukey HSD post-hoc test was conducted to determine which countries differ signi
```

In [79]:

```
# Tukey test for salespeople
tukey_sales = pairwise_tukeyhsd(
    endog=df_top10["Boxes_Shipped"],
    groups=df_top10["Sales_Person"],
    alpha=0.05
)

print(tukey_sales)
```

```
              Multiple Comparison of Means - Tukey HSD, FWER=0.05
=================================================================================
       group1              group2       meandiff p-adj   lower    upper   reject
---------------------------------------------------------------------------------
     Beverie Moffet         Brien Boise -27.5747 0.7613 -77.2785  22.1291  False
     Beverie Moffet        Ches Bonnell -29.5515 0.6933 -79.7888  20.6859  False
     Beverie Moffet Dennison Crosswaite   7.6932    1.0 -43.5312  58.9176  False
     Beverie Moffet        Jan Morforth  13.5569 0.9985 -39.8373  66.9512  False
     Beverie Moffet      Jehu Rudeforth -17.4338  0.988 -69.4344  34.5667  False
     Beverie Moffet    Karlen Mccaffrey  27.3084 0.7976 -23.7079  78.3246  False
     Beverie Moffet       Kelci Walkden -18.3991 0.9757 -67.9338  31.1356  False
     Beverie Moffet      Marney O'Breen   3.7472    1.0 -48.6108  56.1052  False
     Beverie Moffet          Oby Sorrel  -3.1895    1.0  -53.904  47.5249  False
        Brien Boise        Ches Bonnell  -1.9768    1.0 -51.6806   47.727  False
        Brien Boise Dennison Crosswaite  35.2678 0.4541 -15.4334  85.9691  False
        Brien Boise        Jan Morforth  41.1316 0.2888  -11.761  94.0242  False
        Brien Boise      Jehu Rudeforth  10.1408 0.9998 -41.3445  61.6262  False
        Brien Boise    Karlen Mccaffrey   54.883 0.0208   4.3921  105.374   True
        Brien Boise       Kelci Walkden   9.1756 0.9999 -39.8179  58.1691  False
        Brien Boise      Marney O'Breen  31.3219 0.6591 -20.5244  83.1682  False
        Brien Boise          Oby Sorrel  24.3851 0.8759 -25.8009  74.5711  False
       Ches Bonnell Dennison Crosswaite  37.2446 0.3867 -13.9798   88.469  False
       Ches Bonnell        Jan Morforth  43.1084 0.2386 -10.2859  96.5026  False
       Ches Bonnell      Jehu Rudeforth  12.1176 0.9993 -39.8829  64.1182  False
       Ches Bonnell    Karlen Mccaffrey  56.8598 0.0155   5.8436 107.8761   True
       Ches Bonnell       Kelci Walkden  11.1524 0.9994 -38.3823   60.687  False
       Ches Bonnell      Marney O'Breen  33.2987 0.5881 -19.0593  85.6566  False
       Ches Bonnell          Oby Sorrel  26.3619 0.8244 -24.3525  77.0763  False
Dennison Crosswaite        Jan Morforth   5.8638    1.0 -48.4602  60.1878  False
Dennison Crosswaite      Jehu Rudeforth  -25.127 0.8908 -78.0818  27.8278  False
Dennison Crosswaite    Karlen Mccaffrey  19.6152 0.9731 -32.3733  71.6037  False
Dennison Crosswaite       Kelci Walkden -26.0923 0.8301 -76.6277  24.4432  False
Dennison Crosswaite      Marney O'Breen  -3.9459    1.0 -57.2517  49.3598  False
Dennison Crosswaite          Oby Sorrel -10.8827 0.9997 -62.5751  40.8097  False
       Jan Morforth      Jehu Rudeforth -30.9907 0.7454 -86.0472  24.0657  False
       Jan Morforth    Karlen Mccaffrey  13.7514 0.9985 -40.3763  67.8792  False
       Jan Morforth       Kelci Walkden  -31.956  0.655 -84.6897  20.7777  False
       Jan Morforth      Marney O'Breen  -9.8097 0.9999 -65.2039  45.5845  False
       Jan Morforth          Oby Sorrel -16.7465 0.9931 -70.5899  37.0969  False
     Jehu Rudeforth    Karlen Mccaffrey  44.7422 0.1795  -8.0113  97.4956  False
     Jehu Rudeforth       Kelci Walkden  -0.9653    1.0 -52.2873  50.3568  False
     Jehu Rudeforth      Marney O'Breen   21.181 0.9654  -32.871  75.2331  False
     Jehu Rudeforth          Oby Sorrel  14.2443 0.9975 -38.2174  66.7059  False
   Karlen Mccaffrey       Kelci Walkden -45.7075 0.1126 -96.0319    4.617  False
   Karlen Mccaffrey      Marney O'Breen -23.5612 0.9255 -76.6669  29.5446  False
   Karlen Mccaffrey          Oby Sorrel -30.4979 0.6845  -81.984  20.9882  False
      Kelci Walkden      Marney O'Breen  22.1463 0.9395 -29.5378  73.8305  False
      Kelci Walkden          Oby Sorrel  15.2096 0.9941 -34.8089   65.228  False
     Marney O'Breen          Oby Sorrel  -6.9368    1.0 -59.7527  45.8792  False
---------------------------------------------------------------------------------
```

In [80]:
```
# A Tukey HSD post-hoc test was conducted to determine which of the top 10 salespeo
```

In [81]:
```
# The analysis shows that shipment volume (Boxes_Shipped) varies across both countr
```